

ABM BUSINESS LOGIC BIBLE

SECTION 3 ADDENDUM — ADVANCED COGNITION

Foundational Correction (Opportunity Re-Anchoring) + Tier 7 Cognition Modules (M13–M18)

This addendum does two things of different weight. First, a FOUNDATIONAL CORRECTION: the model is re-anchored from the Account to the Opportunity, because a single account is many parallel buying journeys, not one. This is a model change and is treated as the headline. Second, six new ADVANCED COGNITION modules add economic, competitive, simulation, conversation, expert-knowledge, and market-learning intelligence on top of the corrected model. Each module carries a maturity flag — NEAR-TERM (MVP-eligible) or POST-MVP (design now, activate later).

● **NEAR-TERM** — MVP-eligible

● **POST-MVP** — design now, activate later

PART A — FOUNDATIONAL CORRECTION

FOUNDATIONAL CORRECTION — THE OPPORTUNITY RE-ANCHORING

Section 2 anchored the entire model on the Account (ACCT-IDENTITY-005: every signal, score, window, and strategy resolves to exactly one account). Enterprise reality breaks that assumption. A single account is not one decision journey — it is many. SNB is simultaneously an Open Banking opportunity, an API Management opportunity, a Digital Lending opportunity, and an AI Transformation opportunity — each with its own committee, budget, window, incumbent, champion, and timeline. Collapsing them into one account narrative loses real opportunity paths and mis-allocates capital. The correction: the unit of selling is the OPPORTUNITY, not the Account. The Account becomes the relationship CONTAINER; the Opportunity becomes the decision ANCHOR. This is a model change, not an addition — several Section 2/3 rules and both master equations re-resolve from account-level to opportunity-level.

Why this is a correction, not a module

This is deliberately NOT presented as 'just another module.' Building M13-M18 on an account-anchored model and then re-anchoring to opportunities later would force a rebuild of scoring, windows, committees, strategies, allocation, and the master equations. The re-anchoring must be decided FIRST, even though it is the largest single change, because everything else now points at the Opportunity.

ENTITY 38 (NEW CORE): OPPORTUNITY

A distinct buying journey within an account: one initiative-or-need with its own committee, budget, window, incumbent, and timeline. The new decision anchor. An Account contains one or many Opportunities; everything that previously resolved to an Account now resolves to an Opportunity within it.

WHY #1 — Why must the Opportunity become the anchor instead of the Account?

Because the master equations, windows, committees, and strategies are all properties of a SPECIFIC buying journey, not the whole organization. SNB's Open Banking window has nothing to do with its Digital Lending incumbent. Anchoring on the account forces four different journeys to share one score, one window, one strategy — averaging away the very distinctions that drive correct action.

NEW RULE — OPP-ANCHOR-001: The Opportunity is the primary decision anchor. Effective Opportunity and Final Decision Score are computed PER OPPORTUNITY, not per account. Buying_Window, Buying_Committee_Member, Account_Strategy, Initiative, and Vendor_Intelligence re-resolve to an Opportunity. The Account aggregates its Opportunities but is no longer the unit the equations operate on.

WHY #2 — Why keep the Account at all, rather than replace it with Opportunities?

Because relationship capital, Organizational_DNA, market reputation, and connector access are account-level and shared across all its opportunities. A champion's trust, the bank's risk appetite, and prior interaction history span every journey inside the account. The Account remains the CONTAINER for what is genuinely shared; the Opportunity owns what is journey-specific.

NEW RULE — OPP-CONTAINER-001: Account retains account-level state shared across opportunities: Organizational_DNA, Market_Interaction_History, Connector access, relationship capital, and lifecycle. Opportunity inherits these from its parent Account but maintains its own committee, budget, window, incumbent, strategy, and score. Shared-vs-specific is the dividing line.

WHY #3 — Why can an Opportunity NOT simply equal an Initiative (which already exists)?

Because Initiative is the account's internal project as THEY see it; Opportunity is OUR addressable buying journey as WE pursue it. One initiative may contain several Decimal opportunities (Open Banking program → API origination opportunity + SME lending opportunity), and some opportunities span multiple initiatives. They are related but distinct: Initiative is intelligence about them; Opportunity is our pursuit object.

NEW RULE — OPP-INITIATIVE-001: Opportunity links to one or more Initiatives but is not identical to one. An Initiative is account-side reality (intelligence); an Opportunity is Decimal's addressable journey (pursuit). The mapping may be one-to-one, one-initiative-to-many-opportunities, or many-initiatives-to-one-opportunity.

WHY #4 — Why must opportunities within one account be scored and sequenced independently?

Because they compete for different budgets and have different windows and incumbents. Decimal might pursue SNB's Open Banking opportunity aggressively (open window, weak incumbent) while holding its Digital Lending opportunity (entrenched incumbent, closed window). One account, opposite strategies. Independent scoring is what makes that possible.

NEW RULE — OPP-INDEPENDENT-001: Each Opportunity carries its own Three_Score_Record, Buying_Window, and Account_Strategy (now Opportunity_Strategy). Portfolio allocation (M6) ranks OPPORTUNITIES, not accounts, so two journeys in the same account can receive opposite postures and capital levels simultaneously.

WHY #5 — Deepest reason the Opportunity re-anchoring matters?

Because it aligns the system's fundamental unit with how enterprise buying actually works — many parallel journeys per organization. Without it, the system is structurally blind to most of the real pipeline inside a large account, and will repeatedly mis-read one journey's signal as another's. It is the correction that lets the system see the full opportunity surface, not a flattened account average.

NEW RULE — OPP-PRIMACY-001: Account-level aggregation (e.g. 'SNB is HOT') is a derived ROLL-UP of its opportunities, never a primary score. Any decision, score, window, or strategy that previously resolved to an Account MUST now resolve to a specific Opportunity, except for the explicitly account-level shared state in OPP-CONTAINER-001.

OPPORTUNITY — Field Specification

Field	Type	Business Rule	Source
opportunity_id	UUID	Immutable; the new decision anchor	OPP-ANCHOR-001
account_id	UUID	FK to parent Account (container)	OPP-CONTAINER-001
opportunity_name	String	e.g. 'SNB — Open Banking origination'	OPP-ANCHOR-001
linked_initiative_ids	Array[UUID]	One or many Initiatives (not identical)	OPP-INITIATIVE-001
opportunity_type	Enum	OPEN_BANKING / API_MGMT / DIGITAL_LENDING / AI_TRANSFORMATION / ...	OPP-ANCHOR-001
committee_ids / window_id / strategy_id	Array/UUID	Journey-specific (re-resolved from account)	OPP-INDEPENDENT-001
incumbent_vendor_intel_id	UUID	This journey's incumbent (not the account's)	OPP-INDEPENDENT-001
score_record_id	UUID	This Opportunity's Three_Score_Record	OPP-INDEPENDENT-001
economic_profile_id	UUID	FK to M13 Opportunity Economics (deal size etc.)	OPP-ANCHOR-001
lifecycle_status	Enum	PROSPECT / ENGAGED / PILOT / WON / LOST / DORMANT	OPP-ANCHOR-001

What Re-Resolves from Account-Level to Opportunity-Level

These existing rules/equations change anchor. Engineering must apply these before building the cognition modules, or they will be built on the wrong foundation.

Rule / Area	Re-resolution
ACCT-IDENTITY-005 → OPP-ANCHOR-001	Signals/scores/windows/strategies now resolve to an OPPORTUNITY (within an account), not directly to the account.
Master Equations	Effective Opportunity and Final Decision Score are computed PER OPPORTUNITY. Account scores are roll-ups.
WINDOW-* (M5)	Buying_Window belongs to an Opportunity. One account may have several simultaneous windows in different states.
BCM-* (M3/M4)	Buying_Committee_Member belongs to an Opportunity's committee. The same Contact may sit on multiple opportunity committees with different stances.
STRAT-* (M4)	Account_Strategy becomes Opportunity_Strategy — one active strategy per OPPORTUNITY. Account holds the shared relationship posture only.
PORT-* (M6)	Portfolio_Allocation ranks and funds OPPORTUNITIES. Concentration limits apply per account AND per opportunity.
VENDOR-* (M3/M14)	Vendor_Intelligence/incumbency is per Opportunity — different journeys have different incumbents.
DNA / Market_History / Connector (Account)	REMAIN account-level shared state (OPP-CONTAINER-001) — inherited by all opportunities, not duplicated.

M19 — OPPORTUNITY GRAPH ENGINE

● NEAR-TERM (foundational)

Discovers, structures, and maintains the opportunity graph within each account — initiative nodes, stakeholder nodes, vendor nodes, budget nodes, and the influence paths between them. Owns the Opportunity entity and keeps account intelligence decomposed into distinct journeys rather than one flattened narrative.

WHY #1 — Why does the opportunity graph need a dedicated engine rather than living in enrichment (M3)?

Because decomposing an account into distinct opportunities is an ongoing structural judgment, not a one-time data fetch. M3 gathers facts; M19 decides how those facts partition into separate journeys, which committee belongs to which journey, and which budget funds which. That partitioning is the foundation everything else now depends on.

NEW RULE — M19-GRAPH-001: M19 owns construction and maintenance of the per-account opportunity graph: it creates Opportunity entities, assigns committees/windows/incumbents/budgets to the correct Opportunity, and maintains influence paths between nodes. M3 supplies facts; M19 partitions them into journeys.

WHY #2 — Why model stakeholders, vendors, and budgets as NODES rather than fields?

Because the same stakeholder can influence multiple opportunities, the same incumbent can defend several, and budgets can be shared or contested across journeys. A node-and-edge graph captures these many-to-many realities that flat fields on a single Opportunity cannot.

NEW RULE — M19-NODES-001: The opportunity graph contains typed nodes (Opportunity, Stakeholder, Vendor, Budget, Initiative) and influence edges. A stakeholder node may connect to several Opportunity nodes; a budget node may fund or be contested across journeys. Cross-opportunity influence is first-class.

WHY #3 — Why must M19 detect when one signal belongs to a NEW opportunity vs an existing one?

Because mis-assigning a signal to the wrong journey corrupts that journey's score and window. A new Open Banking mandate must spawn (or attach to) the Open Banking opportunity, not inflate the Digital Lending one. M19 performs opportunity-resolution, the journey-level analogue of M1's entity resolution.

NEW RULE — M19-RESOLVE-001: M19 resolves each blessed signal to the correct Opportunity within the account (creating a new Opportunity when none fits). Ambiguous signals attach at account level pending resolution, never mis-assigned to an arbitrary journey. This is opportunity-resolution, parallel to M1 entity-resolution.

WHY #4 — Why does M19 sit upstream of scoring (M2) and strategy (M4) now?

Because M2 and M4 now operate per opportunity — they cannot score or strategize until M19 has established WHICH opportunity the intelligence belongs to. M19 becomes part of the mandatory pipeline: detect → bless → resolve-to-opportunity → score → reason.

NEW RULE — M19-PIPELINE-001: M19 runs after M0 blessing and before M2 scoring in the re-anchored pipeline. Scoring, windows, committees, and strategy all key off the Opportunity M19 assigns. No per-opportunity decision may proceed on intelligence M19 has not yet resolved to a journey.

WHY #5 — Deepest reason M19 exists?

Because it makes the system see the account the way the account actually is — a portfolio of parallel buying journeys — rather than a single averaged narrative. M19 is what turns 'SNB is one deal' into 'SNB is four deals, each at a different stage against a different incumbent.' It is the structural truth the whole re-anchored architecture rests on.

NEW RULE — M19-PRIMACY-001: M19 is the authority on opportunity structure within an account. Account-level roll-ups are derived from M19's graph; no module may treat an account as a single opportunity when M19 has decomposed it into several.

① Workflow Logic (step-by-step)

1. Blessed signal / enrichment arrives for an account (from M0/M3)
2. Opportunity-resolution: does this belong to an existing Opportunity, or spawn a new one?
3. If new → create Opportunity (type, linked initiatives); if existing → attach
4. Assign/Update nodes: stakeholders → committees, vendors → incumbents, budgets → funding
5. Maintain influence edges (cross-opportunity stakeholder/budget links)
6. Emit per-Opportunity events to M2 (score), M5 (window), M4 (strategy), M13 (economics)
7. Maintain account roll-up as a derived view over the opportunity graph

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Signal (blessed) • Enrichment / Initiative / Buying_Committee_Member / Vendor_Intelligence • Organizational_DNA (account context) • Rule_Registry (resolution thresholds)	• Opportunity • opportunity graph nodes/edges • re-assigned committee/window/incumbent links • System_Event (opportunity created/resolved)

③ Decision Nodes & Gates

Decision node	Branch logic
Signal fits an existing Opportunity?	YES → attach; NO → spawn new Opportunity
Opportunity ambiguous?	YES → hold at account level pending resolution (never mis-assign)
Stakeholder spans multiple opportunities?	YES → multi-edge node; NO → single-journey member
Budget shared/contested across journeys?	YES → shared budget node with contention flag

④ **Module Boundary & Orchestration**

M19 owns opportunity STRUCTURE only — partitioning accounts into journeys and maintaining the graph. It does NOT score (M2), time (M5), strategize (M4), or value (M13) opportunities — it establishes WHICH opportunity those modules act on. Orchestration: inserted into the mandatory pipeline after M0, before M2.

PART B — TIER 7: ADVANCED COGNITION MODULES

Six modules built on the re-anchored (opportunity-centric) model. Together they move the system from operational intelligence to economic, competitive, and strategic cognition — the layers of a true Cognitive Enterprise Operating System. Maturity flags indicate MVP sequencing.

Module	Owns	Maturity
M13	Opportunity Economics	● NEAR-TERM
M14	Competitive Intelligence	● NEAR-TERM
M17	Expert Knowledge Extraction	● NEAR-TERM
M15	Strategy Simulation	● POST-MVP
M16	Conversation Intelligence	● POST-MVP
M18	Market Learning	● POST-MVP

M13 — OPPORTUNITY ECONOMICS

● NEAR-TERM (MVP-eligible)

Predicts the economic magnitude of each opportunity — deal size, budget probability, revenue realization, lifetime value — supplying the dollar units the scoring and allocation equations were missing.

WHY #1 — Why does the system need economic modeling when it already scores attractiveness?

Because attractiveness without magnitude is half the picture. The current Effective Opportunity equation ranks fit, intent, timing, and cost but never asks whether the prize is \$50K or \$5M. Two opportunities with identical signals score identically even if one is 100x the value. EROA's numerator has been missing its dollar units.

NEW RULE — M13-VALUE-001: M13 produces an economic profile per Opportunity (estimated_deal_size, budget_probability, procurement_cycle_value, revenue_realization_probability, expected_LTV). Estimated value enters the Effective Opportunity equation as a value term, so magnitude — not just attractiveness — drives ranking.

WHY #2 — Why model budget PROBABILITY, not just budget size?

Because a large budget that may never be approved is worth less than a smaller funded one. Procurement reality is probabilistic — mandates get cut, budgets freeze. Modeling the probability that money actually flows prevents the system from chasing impressive-sounding but unfunded opportunities.

NEW RULE — M13-PROBABILITY-001: M13 separates budget SIZE from budget PROBABILITY (likelihood of approval and release). Expected value = size x probability x realization. A large unfunded opportunity is discounted below a smaller funded one; org instability and reorg states (M-org) lower budget probability.

WHY #3 — Why feed economics into M6 allocation, not just M2 scoring?

Because capital should flow to expected RETURN, and return requires value. M6's EROA was dividing opportunity by cost without a real value numerator. With M13, allocation becomes genuine return-on-capital: a moderate-fit \$5M opportunity can rightly outrank a high-fit \$50K one.

NEW RULE — M13-ALLOCATION-001: M13's expected value is the value numerator in M6's EROA. Allocation maximizes expected economic return under capital constraint, not just attractiveness. Negative or trivial expected-value opportunities are de-prioritized regardless of signal strength.

WHY #4 — Why track realization probability and LTV beyond the initial deal?

Because a first deal that opens a multi-year platform relationship is worth more than a one-off of the same size. Land-and-expand economics and reference value (via the dependency graph) mean true value extends past the initial contract.

NEW RULE — M13-LTV-001: M13 estimates expected lifetime value including expansion potential and reference value (linked to Market_Dependency_Graph). Opportunities that are keystones or platform-entry points carry LTV above their initial deal size; this feeds M6 keystone valuation.

WHY #5 — Deepest reason M13 exists?

Because a system that allocates finite capital without modeling economic value is flying blind on the one dimension that ultimately matters — return. M13 gives the system economic gravity: the sense that not all opportunities deserve equal capital because not all are worth the same. It turns prioritization into investment.

NEW RULE — M13-PRIMACY-001: No opportunity may be allocated significant capital without an M13 economic profile (or an explicit unknown-value flag that caps the capital committed). Value estimates carry confidence; low-confidence estimates cap committed capital until refined.

① Workflow Logic (step-by-step)

1. Opportunity created/updated by M19 → request economic profile
2. Estimate deal size from opportunity type, account size, initiative criticality, comparable deals
3. Estimate budget probability from funding signals, sponsorship, org stability
4. Estimate realization probability + expected LTV (expansion + reference value)
5. Compute expected economic value (size x budget-prob x realization) with confidence
6. Emit value term to M2 (opportunity equation) and M6 (EROA numerator)
7. Refine as new signals/enrichment arrive; lower confidence caps committed capital

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Opportunity (type, linked initiatives) • Account (size, segment) • Initiative (criticality) • Signal (funding/budget) • Organizational_Assessment (stability) • Market_Dependency_Graph (reference value) • Decision_Pattern (comparable deals)	• economic_profile (on Opportunity) • value term → Three_Score_Record • System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Comparable-deal data available?	NO → wider confidence band, cap committed capital; YES → tighter estimate
Budget probability below floor?	YES → discount expected value heavily
Keystone / platform-entry?	YES → add LTV + reference value above initial size
Value confidence low?	YES → cap capital committed until refined (M13-PRIMACY-001)

④ Module Boundary & Orchestration

M13 estimates economic value ONLY. It does NOT score fit/intent (M2), allocate (M6), or assess competition (M14) — it supplies the value numerator the equations were missing. Orchestration: triggered by M19 opportunity events; estimates feed M2/M6.

M14 — COMPETITIVE INTELLIGENCE ENGINE

● NEAR-TERM (MVP-eligible)

Models the competitive battlefield per opportunity — incumbent vulnerability, displacement probability, political protection, replacement triggers — turning a static 'vendor exists' fact into live displacement dynamics.

WHY #1 — Why isn't Vendor_Intelligence (M3) enough?

Because M3 captures the incumbent as a static fact (exists, entrenchment level, renewal date). It does not model whether that incumbent is VULNERABLE — expanding or retreating, satisfied or failing, protected or exposed. The current system sees a vendor; it needs to see a battlefield, because enterprise sales is mostly displacement, not greenfield.

NEW RULE — M14-DYNAMICS-001: M14 maintains competitive dynamics per Opportunity: competitor_strength, displacement_probability, incumbent_satisfaction, replacement_triggers, political_protection_index. These are live, evolving assessments distinct from M3's static Vendor_Intelligence facts.

WHY #2 — Why model political protection specifically?

Because in KSA banking, an incumbent can be technically weak yet politically immovable (a relationship-based or regulator-favored vendor), or technically strong yet politically exposed. Ignoring political protection makes the system pursue technically-winnable but politically-impossible displacements — burning capital on dead ends.

NEW RULE — M14-POLITICAL-001: M14 maintains a political_protection_index per incumbent (relationship depth, regulatory favor, executive sponsorship of the incumbent). High protection suppresses displacement_probability even when technical vulnerability is high; the system avoids politically-impossible displacements.

WHY #3 — Why track replacement triggers rather than just current state?

Because displacement happens at moments — a failed project, a renewal, a leadership change, a public RFP. Knowing the triggers lets the system wait for the opening rather than pushing against a stable incumbent. This is timing-as-weapon applied to competition.

NEW RULE — M14-TRIGGER-001: M14 tracks replacement triggers (incumbent failure, renewal approach, sponsor departure, public RFP, dissatisfaction signals). A trigger firing opens or amplifies a displacement_Buying_Window (via M5) and raises displacement_probability. Absent triggers, entrenched incumbents are de-prioritized.

WHY #4 — Why feed competitive dynamics into strategy (M4) and simulation (M15)?

Because the right strategy depends on the incumbent's likely response. A vulnerable incumbent invites direct displacement; a protected one demands coexistence or patience. M15 stress-tests strategy against incumbent retaliation — which requires M14's model of how the incumbent will react.

NEW RULE — M14-STRATEGY-001: M14 dynamics condition M4 competitive posture (displace/coexist/wait/greenfield) and feed M15 simulation as the incumbent-response model. A strategy assuming an incumbent will not react, against a strong/protected incumbent, is flagged as unrobust.

WHY #5 — Deepest reason M14 exists?

Because enterprise selling is usually a contest against an incumbent, and a system that cannot read the battlefield will pick the wrong fights. M14 gives the system competitive awareness — the difference between selling into a fantasy of empty space and selling into a contested, dynamic, political reality.

NEW RULE — M14-PRIMACY-001: Any displacement strategy must be backed by an M14 competitive assessment. Pursuing displacement without modeling incumbent vulnerability and political protection is prohibited; the default assumption is that every enterprise opportunity is contested.

① Workflow Logic (step-by-step)

1. Opportunity with incumbent identified (from M19/M3)
2. Assess competitor_strength + incumbent_satisfaction from signals/enrichment
3. Compute political_protection_index (relationships, regulatory favor, sponsorship)

4. Identify + monitor replacement_triggers
5. Compute displacement_probability (vulnerability x trigger presence ÷ protection)
6. On trigger firing → notify M5 (open displacement window) + M2 (rescore)
7. Feed dynamics to M4 (posture) and M15 (incumbent-response simulation)

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Opportunity + Vendor_Intelligence (incumbent) • Signal (competitor/dissatisfaction) • Market_Interaction_History • Organizational_DNA (vendor openness) • Market_Learning (M18 competitor moves) • Rule_Registry	• competitive dynamics (per Opportunity) • displacement window trigger → M5 • System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Incumbent present?	NO → greenfield posture; YES → model battlefield
Political protection high?	YES → suppress displacement probability (avoid dead end)
Replacement trigger firing?	YES → open displacement window (M5) + rescore
Displacement strategy without assessment?	Prohibited (M14-PRIMACY-001)

④ Module Boundary & Orchestration

M14 models competition ONLY. It does NOT set final strategy (M4), simulate full scenarios (M15), or own the static incumbent fact (M3) — it supplies live battlefield dynamics. Orchestration: triggered by M19/M3 incumbent events; feeds M4/M5/M15.

M15 — STRATEGY SIMULATION ENGINE

● POST-MVP (design now, activate later)

Stress-tests M4's proposed strategy against branching scenarios (champion, blocker, procurement escalation, incumbent retaliation) before execution, selecting the ROBUST strategy rather than the best-case one.

WHY #1 — Why simulate when M4 already reasons and chooses a strategy?

Because M4 chooses a strategy but never stress-tests it. It decides 'approach the CTO first' without asking what happens if the CTO says no, if procurement intervenes, or if the incumbent retaliates. Reasoning produces a plan; simulation reveals whether that plan survives contact with a reactive buying group.

NEW RULE — M15-SIMULATE-001: Before execution of a significant strategy, M15 runs branching scenarios (champion / blocker / procurement escalation / incumbent retaliation / sponsor departure) against the M4 strategy, computing an expected outcome and a downside for each branch.

WHY #2 — Why select for ROBUSTNESS rather than highest expected outcome?

Because a strategy with the best average but catastrophic downside can be worse than a slightly lower-average strategy that survives all branches. In high-stakes enterprise deals where a blown approach burns the relationship, robustness (good across scenarios) beats fragile optimization.

NEW RULE — M15-ROBUST-001: M15 selects the strategy that is robust across scenarios (strong expected outcome with bounded downside), not merely the highest best-case outcome. A high-mean / catastrophic-downside strategy is rejected in favor of a robust alternative. Downside weighting reflects brand/relationship capital at risk.

WHY #3 — Why must M15's scenarios use real models (DNA, competitive, committee) rather than guesses?

Because a simulation is only as good as its models of how stakeholders and incumbents react. Random scenarios teach nothing. M15 must draw stakeholder responses from Organizational_DNA and committee stances, and incumbent responses from M14 — making the simulation grounded rather than imaginary.

NEW RULE — M15-GROUNDED-001: M15 scenario branches are parameterized by Organizational_DNA (decision style, consensus), Buying_Committee stances, and M14 competitive dynamics. Ungrounded or arbitrary scenarios are not permitted; simulation quality is bounded by the confidence of its input models.

WHY #4 — Why keep M15 advisory until its predictions are calibrated?

Because an unproven simulation that auto-drives strategy could confidently steer the system wrong — the same risk as treating a counterfactual estimate as fact. M15 outputs must be advisory until M11 has validated its scenario predictions against real outcomes.

NEW RULE — M15-CALIBRATE-001: M15 outputs are ADVISORY (labeled estimates) until M11 calibrates scenario predictions against observed outcomes. Until calibrated, M15 may recommend and flag fragile strategies but may not auto-reject an M4 strategy without human review. Calibration lifts this restriction per decision type.

WHY #5 — Deepest reason M15 exists?

Because mature strategy is not choosing the best plan — it is choosing the plan that survives the buyer reacting. M15 moves the system from picking a strategy to pressure-testing it, from optimism to robustness. It is the difference between a planner and a strategist.

NEW RULE — M15-PRIMACY-001: Significant strategies for high-value or high-blast-radius opportunities should pass M15 robustness checking before execution once M15 is calibrated. A strategy flagged fragile proceeds only with human review (M10).

① Workflow Logic (step-by-step)

1. M4 produces a candidate strategy for a significant opportunity
2. Parameterize scenarios from DNA, committee stances, M14 competitive dynamics
3. Run branches: champion / blocker / procurement escalation / incumbent retaliation / sponsor loss
4. Compute expected outcome + downside per branch
5. Score candidate strategy on robustness (mean + bounded downside)
6. If robust → endorse; if fragile → recommend alternative or flag for M10 review
7. Emit results to M11 for prediction calibration against eventual outcome

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Account/Opportunity_Strategy (M4 candidate) • Organizational_DNA • Buying_Committee_Member (stances) • M14 competitive dynamics • Counterfactual_Record (priors) • Rule_Registry	• simulation result (advisory) • fragility flag → M10 • prediction → M11 calibration • System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Input models above confidence floor?	NO → low-confidence sim, advisory only; YES → run
Strategy robust across branches?	YES → endorse; NO → recommend alternative / flag fragile
M15 calibrated for this decision type?	NO → advisory (cannot auto-reject); YES → may gate execution
High blast-radius opportunity?	YES → robustness check expected pre-execution

④ Module Boundary & Orchestration

M15 stress-tests strategy ONLY. It does NOT create strategy (M4), model competition (M14), or execute (M8) — it pressure-tests M4's output. Until calibrated it is advisory. Orchestration: invoked after M4, before M8/M10 for significant opportunities.

M16 — CONVERSATION INTELLIGENCE

● POST-MVP (excluded from MVP per FSD)

A pre-sales layer between M9 and M10 that tracks live conversations — objections, commitments, sentiment shifts, stakeholder emergence — extracting structured intelligence from replies and (later) handling qualification, objection-handling, and scheduling.

WHY #1 — Why is a conversation layer needed between engagement (M9) and handoff (M10)?

Because the current journey assumes a human takes over the moment a positive reply arrives. But a reply is the start of a conversation — objections, questions, commitments, new stakeholders appearing. M16 extracts structured intelligence from that exchange so the eventual human handoff is informed, not cold.

NEW RULE — M16-TRACK-001: M16 processes conversational replies (post-M9) to extract structured signals: objections, commitments, sentiment shifts, and newly-emerged stakeholders. These enrich the Opportunity's committee and feed M10 handoff with conversation context, not just a 'positive reply' flag.

WHY #2 — Why track stakeholder EMERGENCE during conversation?

Because buying committees reveal themselves through dialogue — a CTO loops in security, procurement appears, a new sponsor is cc'd. These emergent stakeholders must be added to the opportunity's committee in real time, or the committee map goes stale exactly when the deal heats up.

NEW RULE — M16-STAKEHOLDER-001: Stakeholders emerging in conversation are added to the Opportunity's Buying_Committee (via M19/M3) with their observed stance. The committee map updates live during active conversations; emergence is a first-class signal that can reshape strategy.

WHY #3 — Why must M16 be scoped to TRACKING before AUTONOMOUS handling at MVP?

Because the FSD explicitly excludes autonomous negotiation from MVP, and autonomously conducting enterprise conversations carries high brand/relationship risk before the system has earned that trust. M16 starts by extracting and surfacing conversation intelligence FOR the human; autonomous handling is a later, trust-gated expansion.

NEW RULE — M16-SCOPE-001: At first activation M16 TRACKS and surfaces conversation intelligence for human handoff; it does NOT autonomously negotiate. Autonomous qualification/objection-handling/scheduling is a later capability, gated by Trust_Capital_Ledger autonomy for the conversation decision type and explicit human enablement.

WHY #4 — Why feed conversation outcomes into learning (M11/M17)?

Because which objections arise and how they resolve is high-value training signal — it teaches the system common objections, effective responses, and where strategies stall. Conversation data sharpens future reasoning, messaging, and the expert heuristics in M17.

NEW RULE — M16-LEARN-001: M16 emits objection/commitment/resolution data to M11 (attribution + pattern learning) and M17 (expert heuristics). Recurring objections inform Decision_Patterns and message strategy; unresolved objection types surface as Capability_Gaps.

WHY #5 — Deepest reason M16 exists?

Because the gap between a positive reply and a closed meeting is where deals are won or lost, and the system currently goes dark there. M16 is the pre-sales intelligence layer — the system's ability to understand a conversation in motion, not just detect that one started.

NEW RULE — M16-PRIMACY-001: Once active, M16 is the authority on conversation-state intelligence between engagement and handoff. It enriches but never bypasses M10 human handoff at MVP; autonomous action requires earned autonomy and is out of MVP scope.

① Workflow Logic (step-by-step)

1. Reply/conversation event from M9 received
2. Extract objections, commitments, questions, sentiment shift
3. Detect emergent stakeholders → add to Opportunity committee (via M19/M3)
4. Update conversation state on the Opportunity
5. Surface structured conversation context to M10 for informed handoff
6. Emit objection/commitment/resolution data to M11 + M17
7. (POST-MVP, trust-gated) handle qualification/objection/scheduling autonomously

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Engagement_Event / reply content (M9) • Opportunity + committee • Contact.persona_profile • Decision_Pattern (objection handling) • Trust_Capital_Ledger (autonomy) • Rule_Registry	• conversation state (on Opportunity) • new Buying_Committee_Member • objection/commitment data → M11/M17 • System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Emergent stakeholder detected?	YES → add to committee live
MVP scope?	TRACK + surface only; autonomous handling blocked until trust-gated
Objection unresolved/novel?	YES → surface as Capability_Gap
Autonomy earned for conversation type?	NO → human handles (M10); YES (post-MVP) → may assist

④ Module Boundary & Orchestration

M16 tracks conversations ONLY at MVP (autonomous negotiation excluded per FSD). It does NOT replace M10 human handoff; it informs it. Sits between M9 and M10. Orchestration: triggered by M9 reply events; autonomous expansion gated by Trust_Capital_Ledger.

M17 — EXPERT KNOWLEDGE EXTRACTION

● NEAR-TERM (MVP-eligible)

Converts accumulated human-reviewer rationales (from M10) into reusable, named heuristics, reviewer playbooks, and candidate rules — institutionalizing expert judgment instead of letting it die in individual review records.

WHY #1 — Why extract heuristics when M10 already stores reviewer rationales?

Because M10 stores rationales as isolated records — a reviewer's 500 decisions over two years contain a coherent expertise that currently accumulates nowhere reusable. M17 mines those rationales for repeated patterns and turns them into named, applicable heuristics. Without it, expert judgment evaporates with each individual decision.

NEW RULE — M17-EXTRACT-001: M17 analyzes Human_Review_Records to detect recurring reviewer patterns and abstracts them into named heuristics (e.g. 'Reviewer-X Heuristic #14: reject engagement when readiness < 60'). Heuristics carry the evidence count and the reviewer they derive from.

WHY #2 — Why attribute heuristics to specific reviewers rather than blend them?

Because reviewers have different expertise and sometimes disagree. Blending them into one anonymous policy loses the signal of WHO is reliable on WHAT, and hides genuine expert disagreement that itself is informative. Attributed heuristics preserve the source of judgment.

NEW RULE — M17-ATTRIBUTE-001: Heuristics are attributed to their source reviewer and decision domain. Conflicting heuristics between reviewers are retained and surfaced (not silently merged), preserving expert disagreement as signal. Reviewer reliability per domain (from M10 outcomes) weights heuristic confidence.

WHY #3 — Why promote stable heuristics toward Decision_Patterns and candidate rules?

Because a heuristic confirmed across many decisions and validated by outcomes is institutional knowledge that should inform the system broadly — not stay locked to one reviewer's queue. Promotion is how individual expertise becomes organizational capability.

NEW RULE — M17-PROMOTE-001: Heuristics stable across many decisions and outcome-validated may be promoted to Decision_Patterns (M11) and proposed as candidate Rule_Registry entries under governance. Promotion requires evidence thresholds; immutable meta-principles are never auto-created this way (RULEREG-IMMUTABLE-001).

WHY #4 — Why feed heuristics back into AI reasoning (M4) and review prioritization?

Because extracted expertise should improve future AI proposals so reviewers see better drafts and review fewer obvious cases. M17 closes the loop: human judgment trains the AI, which raises the floor, which lets humans focus on genuinely hard cases.

NEW RULE — M17-FEEDBACK-001: Validated heuristics inform M4 reasoning (so AI proposals pre-satisfy known reviewer expectations) and M10 review routing (auto-handling cases that established heuristics clearly resolve, escalating genuinely novel ones). This reduces rubber-stamping pressure by removing trivial reviews.

WHY #5 — Deepest reason M17 exists?

Because an organization's hardest-won asset is the judgment of its experts, and a system that lets that judgment die in individual decisions wastes it. M17 institutionalizes expertise — turning what a reviewer knows into what the system knows. It is the human-side mirror of strategic memory.

NEW RULE — M17-PRIMACY-001: M17 converts human judgment into reusable institutional knowledge under governance. Extracted heuristics inform but never silently override explicit rules or current human review; promotion to rules is always governed (parallels PATTERN-PRIMACY-001).

① Workflow Logic (step-by-step)

1. Human_Review_Records accumulate from M10
2. Mine rationales for recurring decision patterns per reviewer + domain
3. Abstract stable patterns into named, attributed heuristics with evidence count
4. Validate heuristics against eventual outcomes (reviewer reliability weighting)
5. Surface conflicting heuristics between reviewers (do not merge)
6. Promote outcome-validated heuristics → Decision_Patterns / candidate rules (governed)
7. Feed validated heuristics to M4 (better proposals) + M10 (review routing)

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
- Human_Review_Record (M10) - Outcome_Attribution (validation) - Decision_Pattern - Rule_Registry - Trust_Capital_Ledger (reviewer reliability)	- Expert heuristic records - Decision_Pattern (promoted) - candidate Rule_Registry entries (governed) - System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Pattern recurs above evidence threshold?	NO → keep observing; YES → form heuristic
Heuristics conflict between reviewers?	Retain both + surface (no silent merge)
Outcome-validated + stable?	YES → eligible for promotion (governed); NO → advisory
Promotion touches immutable principle?	Forbidden — block

④ Module Boundary & Orchestration

M17 institutionalizes reviewer judgment ONLY. It does NOT make live decisions or override current human review (M10) — it extracts and proposes. Promotion to rules is governed. Orchestration: batch/periodic mining of M10 records; outputs feed M4/M10/M11 under governance.

M18 — MARKET LEARNING ENGINE

● POST-MVP (design now, activate later)

Learns from market events Decimal never participated in — competitor wins, public procurement awards, industry announcements, implementation failures — dramatically widening the training signal beyond Decimal's own outcomes.

WHY #1 — Why learn from events Decimal didn't participate in?

Because the market teaches constantly even when Decimal does nothing. When SNB buys a competitor's platform, that event contains timing, stakeholder, vendor, and narrative lessons — all currently ignored because M11 learns only from Decimal's own actions. The richest training signal is the entire market, not just our slice.

NEW RULE — M18-OBSERVE-001: M18 ingests and learns from third-party market events (competitor wins, public procurement awards, industry announcements, implementation successes/failures) that do not involve Decimal. These become learning signal alongside Decimal's own outcomes (M11).

WHY #2 — Why does market learning especially strengthen Tier 6 (M12)?

Because narrative, dependency, and DNA are market-level judgments best learned from market-wide events. A competitor's win at one bank predicts narrative spread and reference effects across the segment; an implementation failure teaches which approaches fail. M18 is the primary external feed into strategic memory.

NEW RULE — M18-MEMORY-001: M18 feeds Market_Narrative (lifecycle evidence), Market_Dependency_Graph (observed cross-account effects from competitor moves), and Organizational_DNA (observed account behavior in deals Decimal wasn't in). It is the main external evidence source for Tier 6 judgment.

WHY #3 — Why must M18 distinguish observed market fact from inferred lesson?

Because a public award is a fact, but 'they won because of X' is an inference. Treating inferences as facts would corrupt learning with speculation. M18 must label confidence and let inferred lessons inform at discounted weight, like counterfactuals.

NEW RULE — M18-CONFIDENCE-001: M18 separates observed market facts (award happened) from inferred lessons (why it happened), labeling the latter as estimates with confidence. Inferred lessons inform learning at discounted weight and never auto-drive decisions, paralleling counterfactual handling (CF-CONFIDENCE-001).

WHY #4 — Why does M18 feed competitive intelligence (M14)?

Because competitor moves observed in the market are direct evidence of competitor strength, expansion, and vulnerability. A competitor winning repeatedly signals strength; losing or failing signals opening. M18 keeps M14's battlefield model current from real market activity.

NEW RULE — M18-COMPETITIVE-001: M18 routes observed competitor wins/losses/failures to M14, updating competitor strength, displacement probability, and replacement triggers from real market outcomes rather than only account-local signals.

WHY #5 — Deepest reason M18 exists?

Because a system that learns only from its own actions learns slowly and narrowly, while the market runs a continuous experiment it could learn from for free. M18 turns the whole market into a teacher — the single biggest accelerant to the system's judgment. It is learning at market scale.

NEW RULE — M18-PRIMACY-001: M18 broadens the learning loop beyond Decimal's actions. All market-derived lessons flow through the same governed, bounded, reversible machinery as internal learning (M11) and inherit the same guardrails; inferred lessons are confidence-discounted and never override observed fact.

① Workflow Logic (step-by-step)

1. Ingest third-party market events (news, procurement awards, announcements, failure reports) via M1/M0
2. Classify event type + extract observed facts (who, what, when)
3. Infer lessons (timing/stakeholder/vendor/narrative) with explicit confidence
4. Route: narrative/dependency/DNA → M12; competitor moves → M14; patterns → M11
5. Apply lessons at confidence-discounted weight through M11's bounded machinery
6. Retain market events for longitudinal trend + Tier 6 training

② Entities Consumed / Written

Reads (consumes)	Writes (produces)
• Third-party market events (via M1/M0) • Market_Narrative / Market_Dependency_Graph / Organizational_DNA • Vendor_Intelligence • Rule_Registry	• market-derived Feedback_Loop_Entries (discounted) • Tier 6 updates via M12 • M14 competitive updates • System_Event

③ Decision Nodes & Gates

Decision node	Branch logic
Event involves Decimal?	NO → M18 market-learning path; YES → M11 internal path
Fact vs inferred lesson?	Fact → full weight; inference → confidence-discounted
Competitor move?	YES → route to M14 battlefield update
Touches immutable principle?	Forbidden — block (inherits M11 guardrails)

④ Module Boundary & Orchestration

M18 learns from external market events ONLY. It does NOT make live decisions; it feeds M11/M12/M14 through governed, bounded, reversible machinery. Inferred lessons are discounted. Orchestration: market events via M1/M0; learning outputs routed to memory/competitive/analytics modules.

ADDENDUM — COMPLETION SUMMARY

The Opportunity re-anchoring corrects the foundation so the model matches enterprise reality — many parallel journeys per account. On that corrected base, six cognition modules add the economic, competitive, and strategic intelligence that distinguish a Cognitive Enterprise Operating System from an ABM platform. Three are MVP-eligible (M13 Economics, M14 Competitive, M17 Expert Knowledge); three are design-now-activate-later (M15 Simulation, M16 Conversation, M18 Market Learning), each gated on models or volume that won't exist at launch.

New core entity	Opportunity (Entity 38) — the new decision anchor
New foundational module	M19 Opportunity Graph Engine
New cognition modules	M13–M18 (6)
Total system modules	20 (M0–M19) + orchestration
5-Whys in this addendum	40
MVP-eligible cognition	M13 · M14 · M17 · M19
Post-MVP cognition	M15 · M16 · M18
Critical sequencing note	Apply Opportunity re-anchoring BEFORE building M13–M18

The system now spans 20 modules and 38 entities across 7 tiers. From here the unbuilt frontier is integration surfaces (Section 4) and the consolidated master specification.